

Write-Search-Cancel: A Structural Theory of Real-Time Literal-Demand Normalization

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Abstract

We study the structural arithmetic behind real-time literal-demand normalization for the weighted balance languages

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}, \quad \gcd(p,q) = 1.$$

The starting point is the unique minimum nonnegative completion demand associated with each input prefix. We show that every prefix determines a unique dominating kernel level κ , and that the semantic update on one input symbol factors uniquely into a unit cancellation and, when necessary, one primitive zero-residual kernel carry $K = (q,p)$. This yields a realization-independent semantic chronology.

We then separate semantic arithmetic from one-top LIFO representation. A top-rewrite suffix criterion gives an exact test for whether a canonical semantic successor can be realized in one real-time pushdown transition. It exposes two distinct obstruction families: buried cancellation and carry-induced order obstruction. The first forces a unit exchange defect under a natural kernel-economy condition; the second can preserve literal counts while necessarily creating stack-order debt. This motivates a three-channel view of noncanonicity: quantity debt, order debt, and control/sheet debt. We further show that these three channels are independent at the configuration level: each can be nonzero while the other two remain canonical.

For the phase-coherent, kernel-economical WSC count-shadow class, quantity defects lie in the lattice generated by $h = (1, -1)$ and $K = (q, p)$. Every physical transition then admits a unique arithmetic factorization

$$\Delta P = -e_a - \alpha h + MK,$$

where $-e_a$ is the semantic cancellation component, $-\alpha h$ is the SEARCH/exchange component in the abelian count shadow, and MK is physical kernel materialization. A physical materialization level λ separates semantic WRITE generation from eager or lazy physical WRITE chronology and yields a global materialization budget. In the historical 2:1 specialization, the old state potentials become SEARCH-sheet coordinates, the physical length changes $-1, +2, +5$ become materialization amounts $0, 1, 2$, and the identity $b + 2c = n$ follows as a structural corollary.

Finally, we formalize semantic uniqueness: the canonical demand dynamics is unique, whereas physical normalization lifts need not be. The WSC lattice is explicitly scoped as an abelian/Parikh shadow of the full LIFO representation fiber, not as the full fiber itself. The resulting theory is intended as a structural layer toward a broader theory of real-time pushdown normalization.

1 Introduction

A deterministic pushdown construction can be correct for a language while still hiding several logically different mechanisms inside its transition table. In the weighted balance setting, three

operations repeatedly appear: WRITE, SEARCH, and CANCEL. At first sight they look like engineering choices. A WRITE transition pushes a block, a SEARCH transition changes the representation so that a buried demand can be exposed, and a CANCEL transition removes a matched demand. The purpose of this paper is to show that this description is too operational. The three roles arise from a deeper separation between *semantic demand dynamics* and *real-time LIFO realization*.

The central semantic object is the unique minimum nonnegative completion demand. Given a prefix w , it asks for the shortest multiset of future zeros and ones that can complete w to the weighted balance relation. This object is independent of any particular pushdown machine. Its arithmetic is governed by the primitive kernel packet

$$K = (q, p),$$

which carries zero residual. The first main result of the paper is that each prefix count point is dominated by a unique first point on the kernel ray, and that the slack to that point is exactly the shortest completion demand. This produces a unique semantic chronology.

The second layer is representational. A real-time one-top pushdown machine may be unable to implement the canonical semantic update in one input-consuming transition, because the literal cell that must change is buried below the top. This is a locality obstruction, not a semantic ambiguity. The correct response is therefore not to alter the semantic target, but to move to a noncanonical representation of the same semantic object. We show that noncanonicity has more than one channel: literal counts can be wrong relative to the canonical demand, literal order can be wrong even when counts are exact, and finite control can carry additional sheet information.

The third layer is a scoped WSC calculus. In a phase-coherent count-shadow policy, quantity defects form a rank-two integer lattice with primitive directions

$$h = (1, -1), \quad K = (q, p).$$

The first direction is the primitive length-preserving exchange direction; the second is the primitive residual-preserving kernel direction. This gives a unique arithmetic decomposition of each physical count change. The decomposition does not claim that every syntactic transition is literally one SEARCH, one CANCEL, or one WRITE instruction. Rather, it identifies the three structural components carried by a transition.

A fourth theme is uniqueness. The semantic path is unique; the physical lift is not. Eager and lazy materialization can realize the same semantic carry chronology with different physical timing. More generally, the count-shadow lattice does not record stack order or control sheets, so full physical classification requires a richer lift theory. This paper includes the semantic uniqueness results needed to state the structural picture correctly, while deliberately leaving minimal-lift classification and confluence questions to a subsequent study.

Contributions. The paper establishes the following.

- (i) A closed-form dominating-kernel formula for the unique shortest completion demand.
- (ii) A unique semantic CANCEL/carry factorization for every input step.
- (iii) A top-rewrite suffix criterion and an exact obstruction classification for the fixed canonical orientation 1^*0^* .
- (iv) A proof that buried cancellation forces a unit SEARCH exchange under kernel economy, together with a distinct carry-order obstruction that can have zero quantity defect.
- (v) A separation of quantity, order, and control/sheet debt.

- (vi) A phase-coherent WSC defect lattice and a unique physical count-shadow factorization

$$\Delta P = -e_a - \alpha(1, -1) + M(q, p).$$

- (vii) A physical materialization level, eager/lazy dichotomy, backlog law, and global materialization budget.
- (viii) A semantic quotient/uniqueness theorem and an explicit statement of physical lift non-uniqueness.
- (ix) A structural reconstruction of the historical 2:1 state potentials, mod-3 clock, $-1, +2, +5$ stack actions, and $b + 2c = n$ identity.

Scope. The strongest count-shadow theorems below are stated for a clearly defined phase-coherent, kernel-economical realization class. We do *not* claim that every correct DPDA belongs to that class. We also do not identify the WSC lattice with the full LIFO representation fiber. Full canonicity requires correct counts, correct stack order, and the canonical control/sheet state. Finally, uniqueness of a canonical semantic endpoint is not asserted to imply confluence of arbitrary physical schedules.

Related work and novelty boundary. Right-residual and congruence methods are classical in automata theory. In the visibly pushdown setting, Alur, Kumar, Madhusudan, and Viswanathan developed language congruences and canonical automata, while also showing that canonicity and minimality need not coincide in general [1]. Later work established that minimization of visibly pushdown automata is NP-complete [2]. For unrestricted deterministic pushdown automata, language equivalence is decidable by the work of Sénizergues and subsequent simplifications [3, 4]; those results concern behavioral equivalence rather than uniqueness of a physical stack representation. Carayol and Serre study the marking of outgoing edges lying on shortest paths in deterministic real-time pushdown graphs [5]; the minimum-completion arithmetic below is a family-specific structural analysis and is not claimed as a new general shortest-path formalism. Accordingly, the contribution claimed here is the exact WSC decomposition for the weighted-balance normalization family: the semantic CANCEL/carry chronology, the one-top obstruction/debt split, and the phase-coherent physical materialization calculus under the stated policy assumptions.

2 Weighted balance and minimum completion demand

Fix coprime positive integers p, q and define

$$L_{p,q} = \{w \in \{0, 1\}^* : qN_1(w) = pN_0(w)\}.$$

We use the residual convention

$$\rho(w) = qN_1(w) - pN_0(w). \tag{1}$$

A completion demand is a pair $(x, y) \in \mathbb{N}_0^2$, where x zeros and y ones are appended. The completion condition is

$$px - qy = \rho(w). \tag{2}$$

The primitive zero-residual direction is

$$K = (q, p), \tag{3}$$

because $pq - qp = 0$.

Lemma 2.1 (Kernel-fiber lemma). *If two integer demand pairs represent the same residual, then their difference is an integer multiple of K . More precisely, if*

$$px - qy = px' - qy',$$

then

$$(x, y) - (x', y') = t(q, p)$$

for some $t \in \mathbb{Z}$.

Proof. The equality gives

$$p(x - x') = q(y - y').$$

Since $\gcd(p, q) = 1$, $q \mid x - x'$ and $p \mid y - y'$. Thus $x - x' = tq$ and $y - y' = tp$ for the same integer t . $\square$

Definition 2.2 (Minimum completion demand). *For a residual ρ , let $\mu(\rho) = (x, y)$ be the nonnegative solution of (2) minimizing $x + y$. For a prefix w , write $\mu(w) = \mu(\rho(w))$.*

The following formula is the semantic backbone of the paper.

Theorem 2.3 (Dominating kernel level). *Let*

$$a = N_0(w), \quad b = N_1(w).$$

Define

$$\kappa(w) = \max \left(\left\lceil \frac{a}{q} \right\rceil, \left\lceil \frac{b}{p} \right\rceil \right). \quad (4)$$

Then the unique minimum nonnegative completion demand is

$$\mu(w) = (q\kappa(w) - a, p\kappa(w) - b). \quad (5)$$

Equivalently,

$$(a, b) + \mu(w) = \kappa(w)K. \quad (6)$$

Proof. If (x, y) completes w , then

$$q(b + y) = p(a + x).$$

Coprimality implies the existence of an integer k such that

$$a + x = qk, \quad b + y = pk.$$

Hence

$$x = qk - a, \quad y = pk - b.$$

Nonnegativity is equivalent to

$$k \geq \left\lceil \frac{a}{q} \right\rceil, \quad k \geq \left\lceil \frac{b}{p} \right\rceil.$$

Thus the least admissible value is exactly $\kappa(w)$. The completion length is

$$x + y = (p + q)k - (a + b),$$

which is strictly increasing in k . Therefore the least admissible k gives the unique minimum completion demand, yielding (5) and (6). $\square$

Remark 2.4 (Geometric interpretation). *The count point (a, b) moves in $\mathbb{N}_0^2$. The kernel ray consists of points $kK = (kq, kp)$. The number $\kappa(w)$ is the first kernel level whose point dominates (a, b) coordinatewise, and $\mu(w)$ is the slack vector from (a, b) to that point.*

Let

$$e_0 = (1, 0), \quad e_1 = (0, 1), \quad r = p + q.$$

Theorem 2.5 (Unique semantic CANCEL-carry factorization). *If the next input symbol is $a \in \{0, 1\}$, then*

$$\mu_{t+1} = \mu_t - e_a + \chi_t K, \tag{7}$$

where

$$\chi_t = \kappa_{t+1} - \kappa_t \in \{0, 1\}.$$

If $\mu_t = (x, y)$, then $\chi_t = 1$ on input 0 iff $x = 0$, and $\chi_t = 1$ on input 1 iff $y = 0$.

Proof. Reading one symbol increases exactly one prefix count by one, so each ceiling in (4) changes by at most one and therefore $\kappa_{t+1} - \kappa_t \in \{0, 1\}$. Equation (6) before and after the input gives

$$\begin{aligned} (N_0, N_1) + \mu_t &= \kappa_t K, \\ (N_0, N_1) + e_a + \mu_{t+1} &= \kappa_{t+1} K. \end{aligned}$$

Subtracting yields (7). If input 0 arrives, the zero slack is $x = q\kappa_t - N_0$. The kernel level rises exactly when the current zero coordinate is already saturated, namely $x = 0$. The input-1 statement is symmetric. $\square$

Corollary 2.6 (Semantic WRITE count). *For every prefix w , the number of semantic kernel carries is*

$$W_{\text{sem}}(w) = \kappa(w).$$

If $w \in L_{p,q}$ has $N_0(w) = qn$ and $N_1(w) = pn$, then

$$W_{\text{sem}}(w) = n.$$

Proof. The empty prefix has $\kappa = 0$, and every semantic carry increments κ by one. For an accepted word, (4) gives $\kappa = n$. $\square$

Corollary 2.7 (Demand-length clock). *At prefix length t ,*

$$|\mu_t| = r\kappa_t - t, \tag{8}$$

and hence

$$|\mu_t| \equiv -t \pmod{r}.$$

Proof. Summing the coordinates in (5) gives

$$|\mu_t| = q\kappa_t - N_0 + p\kappa_t - N_1 = r\kappa_t - t.$$

$\square$

Corollary 2.8 (Semantic chronology uniqueness). *For a fixed input word, the entire sequences (ρ_t) , (κ_t) , (μ_t) , and (χ_t) are uniquely determined by the input and are therefore independent of the physical pushdown realization.*

3 Real-time literal-demand lifts

We now separate the semantic demand system from its pushdown realization.

Definition 3.1 (Exact literal-demand lift). *An exact real-time literal-demand lift consists of a reachable configuration set $\mathcal{C}$, deterministic input transitions δ_a for $a \in \{0, 1\}$, a semantic projection*

$$\Theta : \mathcal{C} \rightarrow \mathcal{M}_{p,q},$$

where $\mathcal{M}_{p,q} = \{\mu(\rho) : \rho \in \mathbb{Z}\}$, and a canonical section

$$\sigma : \mathcal{M}_{p,q} \rightarrow \mathcal{C}$$

such that

$$\Theta \circ \sigma = \text{id}$$

and

$$\Theta(\delta_a(c)) = U_a(\Theta(c)), \quad (9)$$

where U_a is the semantic update from Theorem 2.5.

We take the canonical literal encoding to be

$$\eta(x, y) = 1^y 0^x Z_0, \quad (10)$$

with the stack top on the left and Z_0 a fixed bottom marker during binary-input processing.

Definition 3.2 (One-top real-time locality). *A literal stack transition is one-top if, on a nonempty literal stack, it has the form*

$$XT \longrightarrow \gamma T,$$

where $X \in \{0, 1\}$ is the current top symbol, T is the untouched suffix, and $\gamma \in \{0, 1\}^*$. When the literal stack is empty, a bottom-marker transition may write a literal word above Z_0 . No ε -repair sequence is inserted between binary input symbols.

4 The locality obstruction

Theorem 4.1 (Top-rewrite suffix criterion). *Suppose a one-top transition starts from XT . A desired target literal word S^* can be obtained in that single transition if and only if T is a suffix of S^* .*

Proof. Every one-top successor has the form γT , so direct realization implies that T is a suffix of the target. Conversely, if $S^* = \gamma T$ for some γ , replacing X by γ yields the target in one transition. $\square$

The criterion gives a complete obstruction classification for the fixed orientation (10).

Theorem 4.2 (Canonical direct-realizability classification). *Let the canonical literal demand be $1^y 0^x Z_0$.*

- (a) *Input 1 is directly canonical-realizable for every canonical demand.*
- (b) *Input 0 fails to be directly canonical-realizable exactly in the following two families:*
 - (i) $x > 0$ and $y > 0$ (buried-CANCEL obstruction);
 - (ii) $x = 0$, $y \geq 2$, and $q \geq 2$ (carry-order obstruction).

The reverse block orientation gives the symmetric statement with 0 and 1 exchanged.

Proof. For input 1, if $y > 0$, the canonical target is obtained by popping the top 1. If $y = 0$ and $x > 0$, the semantic target is

$$1^{p-1}0^{x+q}Z_0.$$

The untouched suffix after the top 0 is $0^{x-1}Z_0$, which is a suffix of the target; one may replace the top 0 by $1^{p-1}0^{q+1}$. If $x = 0$, the same target is written above the bottom marker.

For input 0, if $y = 0$ and $x > 0$, direct cancellation pops the top 0. If $x > 0$ and $y > 0$, the semantic target is $1^y0^{x-1}Z_0$, while the untouched suffix is $1^{y-1}0^xZ_0$, which is not a suffix of the target. This is the buried-CANCEL obstruction.

It remains to consider $x = 0$. The semantic carry target is

$$1^{y+p}0^{q-1}Z_0.$$

If $y = 0$, a bottom-marker write is direct. If $y = 1$, the untouched literal suffix is empty and the top 1 can be replaced by the entire target literal block. If $y \geq 2$, the untouched suffix is $1^{y-1}Z_0$. This is a suffix of the target exactly when $q - 1 = 0$, namely $q = 1$. Thus the second obstruction family is exactly $x = 0, y \geq 2, q \geq 2$. $\square$

5 Quantity, order, and control debt

Let a rich configuration c have literal stack word $S(c)$ above the bottom marker. Define the physical count vector

$$P(c) = \text{Par}(S(c)) = (N_0(S(c)), N_1(S(c))).$$

Definition 5.1 (Quantity defect). *The quantity defect is*

$$Q(c) = \Theta(c) - P(c). \quad (11)$$

Canonical configurations have $Q = 0$, but the converse is false because the Parikh map forgets order.

Proposition 5.2 (Universal representation conservation). *For every exact input transition on symbol a ,*

$$\Delta P + \Delta Q = -e_a + \chi K. \quad (12)$$

Proof. By definition $Q = \Theta - P$, so $\Delta Q = \Delta\Theta - \Delta P$. The semantic change is Theorem 2.5. $\square$

Definition 5.3 (Order defect). *For a literal word $S = S_1 \cdots S_m$, define*

$$\Omega(S) = \#\{(i, j) : 1 \leq i < j \leq m, S_i = 0, S_j = 1\}. \quad (13)$$

Lemma 5.4 (Block-order test). $\Omega(S) = 0$ if and only if $S \in 1^*0^*$.

Proof. If $S \in 1^*0^*$, no 0 occurs before a later 1, so $\Omega = 0$. Conversely, if a word is not in 1^*0^* , some 0 is followed later by a 1, producing an inversion. $\square$

A third channel is finite control. A configuration can have the correct literal counts and canonical literal order while still being in a noncanonical control/sheet state. Thus full canonicity is a conjunction of three conditions.

Proposition 5.5 (Three-channel canonicity). *For the fixed literal encoding (10), a rich configuration is canonical exactly when:*

- (i) its quantity defect is zero;

(ii) its literal order defect is zero;

(iii) its control state is the canonical section state.

Proof. The forward implication is immediate. Conversely, $Q = 0$ makes the physical literal counts equal to the semantic demand counts; $\Omega = 0$ forces the literal word into the block order 1^*0^* ; the fixed counts then identify that word with $\eta(\Theta(c))$. The final condition selects the canonical control representative. $\square$

Proposition 5.6 (Configuration-level separation of the three defect channels). *At the level of exact semantic representations used in this theory, none of the three canonicity channels is implied by the other two. Each channel admits a witness for which the other two are canonical.*

Proof. Quantity-only defect. In the historical 2:1 schedule interaction analyzed by the companion lift theory [6], the kernel-excess configuration $(C, 1110)$ is compared with the canonical representative $(C, 1)$ of the same semantic target. The control sheet and block order are canonical, while the physical count differs from the minimum demand by one kernel packet.

Order-only defect. The carry-order repair of Theorem 5.9 preserves the exact canonical Parikh counts, so $Q = 0$, but Theorem 5.10 gives positive inversion count whenever $y \geq 2$ and $q \geq 2$.

Control-only defect. Adjoin to any exact literal lift an auxiliary two-sheet control bit that is ignored by the stack dynamics, semantic projection, and acceptance, but toggles on every consumed input symbol. A balanced input of odd length in the 2:1 system, for example 011, returns the base machine to its canonical semantic stack while leaving the auxiliary sheet noncanonical. Thus quantity and order are canonical while the control channel is not.

Therefore quantity, order, and control/sheet canonicity are independent physical requirements. The last witness belongs to an enlarged exact-lift presentation rather than to the fixed historical three-state table; the proposition concerns independence of the channels in the representation theory, not independence inside one particular finite control graph. $\square$

5.1 Buried cancellation forces a unit exchange under economy

Definition 5.7 (Kernel economy). *A literal-demand representation is kernel-economical if*

$$|P(c)| \leq |\Theta(c)| \quad (14)$$

for every reachable configuration.

This condition prohibits a realization from adding complete zero-residual kernel packets merely as redundant workspace.

Theorem 5.8 (Forced unit SEARCH for buried cancellation). *Assume one-top real-time locality and kernel economy. Start from a canonical mixed demand*

$$\Theta(c) = (x, y), \quad x > 0, y > 0,$$

encoded as $1^y 0^x Z_0$. On input 0, the one-step successor has quantity defect

$$Q' = (-1, 1) = -(1, -1). \quad (15)$$

Thus the count shadow undergoes a forced unit SEARCH exchange. The reverse canonical orientation gives the opposite sign.

Proof. The semantic target is $(x - 1, y)$, of total length $x + y - 1$. A one-top transition can only replace the blocking top 1 by some word γ . The successor physical literal length is

$$x + y - 1 + |\gamma|.$$

Kernel economy requires this to be at most the semantic target length $x + y - 1$, so $|\gamma| = 0$. Hence the top 1 must be popped. The physical successor counts are $(x, y - 1)$, while the semantic target is $(x - 1, y)$. Their difference is $(-1, 1)$. $\square$

5.2 Carry obstruction can be purely order-theoretic

Theorem 5.9 (Count-exact carry-order repair). *Assume $x = 0$, $y \geq 2$, and $q \geq 2$, and read input 0 from the canonical word $1^y Z_0$. The semantic target is*

$$1^{y+p} 0^{q-1} Z_0.$$

A one-top transition can preserve the target counts exactly by replacing the top 1 with

$$\gamma = 1^{p+1} 0^{q-1}.$$

The resulting successor has quantity defect zero but nonzero order defect.

Proof. The successor word is

$$1^{p+1} 0^{q-1} 1^{y-1} Z_0.$$

It contains $y + p$ ones and $q - 1$ zeros, exactly the semantic target counts, so $Q' = 0$. Because $q - 1 > 0$ and $y - 1 > 0$, at least one 0 precedes a later 1, so the word is noncanonical. $\square$

Theorem 5.10 (Minimum carry-order debt). *Under the hypotheses of Theorem 5.9, among all one-top successors with exact target counts, the minimum order defect is*

$$\Omega_{\min} = (q - 1)(y - 1). \quad (16)$$

Proof. The untouched suffix is $1^{y-1} Z_0$. Exact target counts force the replacement word to contain exactly $q - 1$ zeros and $p + 1$ ones. Every one of those $q - 1$ zeros occurs before all $y - 1$ untouched suffix ones, producing at least $(q - 1)(y - 1)$ inversions. The replacement $1^{p+1} 0^{q-1}$ has no additional inversions inside the replacement itself and therefore attains the bound. $\square$

Remark 5.11 (Obstruction-debt principle). *Theorems 5.8 and 5.9 show why quantity defect must not be identified with full noncanonicity. A LIFO obstruction forces departure from the canonical section, but the resulting normalization debt may be stored in quantity, order, finite control, or a combination of these channels.*

6 The phase-coherent WSC quantity lattice

The preceding section was representation-general within the literal-demand setting. We now isolate a count-shadow policy class in which the WRITE-SEARCH-CANCEL arithmetic becomes especially rigid.

Set

$$h = (1, -1), \quad K = (q, p), \quad r = p + q.$$

Definition 6.1 (Phase coherence). *A reachable configuration is phase-coherent if*

$$|P(c)| \equiv |\Theta(c)| \pmod{r}. \quad (17)$$

A realization is phase-coherent if every reachable configuration is phase-coherent.

Theorem 6.2 (WSC defect lattice). *Define*

$$\mathcal{D}_{p,q} = \mathbb{Z}h + \mathbb{Z}K.$$

Then

$$\mathcal{D}_{p,q} = \{(u, v) \in \mathbb{Z}^2 : u + v \equiv 0 \pmod{r}\}. \quad (18)$$

Consequently, every phase-coherent quantity defect has a unique decomposition

$$Q = sh + wK, \quad s, w \in \mathbb{Z}. \quad (19)$$

Proof. Both generators satisfy the congruence: $h_0 + h_1 = 0$ and $K_0 + K_1 = q + p = r$. Hence $\mathcal{D}_{p,q}$ is contained in the right side of (18).

Conversely, let $Q = (u, v)$ satisfy $u + v \equiv 0 \pmod{r}$. Put

$$w = \frac{u + v}{r}, \quad s = \frac{pu - qv}{r}. \quad (20)$$

The first number is integral by assumption. Since $p \equiv -q \pmod{r}$,

$$pu - qv \equiv -q(u + v) \equiv 0 \pmod{r},$$

so $s \in \mathbb{Z}$. Direct substitution gives $Q = sh + wK$. Uniqueness follows because h and K are linearly independent; their determinant is $p + q = r \neq 0$. $\square$

Define the scalar maps

$$\Phi(Q) = pQ_0 - qQ_1, \quad \Lambda(Q) = Q_0 + Q_1. \quad (21)$$

Corollary 6.3 (Defect coordinates). *For $Q = sh + wK$,*

$$\Phi(Q) = rs, \quad \Lambda(Q) = rw. \quad (22)$$

Hence

$$s = \frac{\Phi(Q)}{r}, \quad w = \frac{\Lambda(Q)}{r}.$$

Proof. One has $\Phi(h) = p + q = r$, $\Phi(K) = 0$, $\Lambda(h) = 0$, and $\Lambda(K) = r$. $\square$

Corollary 6.4 (Defect quotient).

$$\mathbb{Z}^2 / \mathcal{D}_{p,q} \cong \mathbb{Z}_r.$$

Proof. The homomorphism $(u, v) \mapsto u + v \pmod{r}$ is surjective and has kernel $\mathcal{D}_{p,q}$ by Theorem 6.2. $\square$

Remark 6.5 (WSC lattice is an abelian shadow). *The map $S \mapsto \text{Par}(S)$ forgets literal order, and Q also forgets the finite control state. Therefore $\mathcal{D}_{p,q}$ is not the full representation fiber. It is the abelian/Parikh count shadow of that fiber. In particular, two configurations can have the same (s, w) coordinates and still have different stack words or control sheets.*

Corollary 6.6 (Kernel economy in lattice coordinates). *For a phase-coherent configuration, kernel economy (14) is equivalent to*

$$w \geq 0.$$

Proof. Since $|\Theta| - |P| = \Lambda(Q) = rw$, the two conditions are equivalent. $\square$

7 Physical materialization and the WSC transition factorization

Let $Q_t = s_t h + w_t K$ along a phase-coherent run.

Definition 7.1 (Physical materialization level). *Define*

$$\lambda_t = \kappa_t - w_t. \quad (23)$$

Theorem 7.2 (Physical length formula). *At prefix length t ,*

$$|P_t| = r\lambda_t - t. \quad (24)$$

Proof. By definition,

$$|P_t| = |\Theta_t| - \Lambda(Q_t).$$

Corollary 2.7 gives $|\Theta_t| = r\kappa_t - t$, while Corollary 6.3 gives $\Lambda(Q_t) = rw_t$. Substitution yields (24). $\square$

Definition 7.3 (Physical WRITE amount). *For one transition, define*

$$M_t = \lambda_{t+1} - \lambda_t. \quad (25)$$

Theorem 7.4 (Materialization amount from stack length). *For every phase-coherent one-top transition,*

$$M_t = \frac{\Delta|P_t| + 1}{r}. \quad (26)$$

In particular, $M_t \in \mathbb{N}_0$.

Proof. Subtract (24) at times $t + 1$ and t :

$$\Delta|P_t| = r(\lambda_{t+1} - \lambda_t) - 1 = rM_t - 1.$$

The phase-coherent lattice coordinates make λ_t integral, hence M_t is integral. A one-top rewrite can delete at most the current top literal cell, hence $\Delta|P_t| \geq -1$, and therefore $M_t \geq 0$. $\square$

Theorem 7.5 (Backlog law). *The deferred-kernel coordinate satisfies*

$$w_{t+1} - w_t = \chi_t - M_t. \quad (27)$$

Equivalently, with cumulative semantic arrivals $A_t = \kappa_t$ and cumulative physical materializations $B_t = \lambda_t$,

$$w_t = A_t - B_t. \quad (28)$$

Proof. From $w = \kappa - \lambda$,

$$\Delta w = \Delta \kappa - \Delta \lambda = \chi - M.$$

The cumulative form is the definition of A_t, B_t . $\square$

Theorem 7.6 (Canonical carry dichotomy). *Assume a phase-coherent run and kernel economy. Start from a canonical quantity state, so $w_t = 0$.*

(a) *If $\chi_t = 0$, then $M_t = 0$.*

(b) *If $\chi_t = 1$, then $M_t \in \{0, 1\}$.*

Thus a canonical semantic carry is either eagerly materialized or exactly one kernel packet is deferred.

Proof. By Corollary 6.6, $w_{t+1} \geq 0$. From (27),

$$w_{t+1} = \chi_t - M_t.$$

Since $M_t \in \mathbb{N}_0$ and $\chi_t \in \{0, 1\}$, the stated possibilities are the only ones. $\square$

Let

$$\alpha_t = s_{t+1} - s_t. \quad (29)$$

Theorem 7.7 (Unique physical WSC count-shadow factorization). *Every phase-coherent one-top transition on input a satisfies*

$$\boxed{\Delta P = -e_a - \alpha_t h + M_t K.} \quad (30)$$

The coefficients $\alpha_t \in \mathbb{Z}$ and $M_t \in \mathbb{N}_0$ are unique.

Proof. The defect change is

$$\Delta Q = \alpha_t h + (w_{t+1} - w_t)K = \alpha_t h + (\chi_t - M_t)K$$

by Theorem 7.5. Substitute this into the universal conservation law (12):

$$\Delta P = -e_a + \chi_t K - \alpha_t h - (\chi_t - M_t)K = -e_a - \alpha_t h + M_t K.$$

Uniqueness follows from linear independence of h and K , together with the fixed semantic term $-e_a$. $\square$

Corollary 7.8 (Reconstruction of WSC charges). *The physical materialization and SEARCH coordinates are recovered from the physical count effect by*

$$M = \frac{\Delta|P| + 1}{r}, \quad (31)$$

and

$$\alpha = -\frac{p(\Delta P_0 + e_{a,0}) - q(\Delta P_1 + e_{a,1})}{r}. \quad (32)$$

Proof. Equation (31) is Theorem 7.4. Apply the residual projection $(u, v) \mapsto pu - qv$ to

$$\Delta P + e_a = -\alpha h + MK.$$

Since $\Phi(h) = r$ and $\Phi(K) = 0$, one obtains (32). $\square$

Remark 7.9 (Meaning of the factorization). *Equation (30) is a role decomposition, not a syntactic transition taxonomy. The term $-e_a$ is the semantic cancellation forced by the input. The h -component measures the SEARCH/exchange effect in the abelian quantity shadow. The K -component measures how many kernel packets are physically materialized in that transition. A single concrete transition may combine these roles.*

Corollary 7.10 (General physical phase clock). *Every phase-coherent run satisfies*

$$|P_t| \equiv -t \pmod{r}.$$

Proof. Immediate from (24). $\square$

Theorem 7.11 (Global materialization budget). *Assume a phase-coherent run, kernel economy, and start from the canonical empty configuration. Let an accepted word satisfy*

$$N_0 = qn, \quad N_1 = pn.$$

Then

$$\sum_t M_t = n. \quad (33)$$

This equals the total semantic WRITE count $\sum_t \chi_t = n$.

Proof. The input length is rn . At acceptance, the semantic minimum demand is zero. Kernel economy forces $|P| \leq 0$, hence $|P| = 0$. Equation (24) at time rn gives

$$0 = r\lambda_{rn} - rn,$$

so $\lambda_{rn} = n$. Since $\lambda_0 = 0$, telescoping gives

$$\sum_t M_t = \lambda_{rn} - \lambda_0 = n.$$

The semantic equality follows from Corollary 2.6. $\square$

Corollary 7.12 (Deferral capacity bound). *Along a phase-coherent kernel-economical run, at prefix length t ,*

$$\left\lceil \frac{t}{r} \right\rceil \leq \lambda_t \leq \kappa_t, \quad (34)$$

and hence

$$0 \leq w_t \leq \kappa_t - \left\lceil \frac{t}{r} \right\rceil. \quad (35)$$

Proof. Physical nonnegativity and (24) imply $r\lambda_t - t \geq 0$, giving the lower bound. Kernel economy gives $w_t \geq 0$, so $\lambda_t = \kappa_t - w_t \leq \kappa_t$. Equation (35) follows. $\square$

Proposition 7.13 (Abstract count-shadow fiber region). *Let $\Theta(c) = \mu = (x, y)$ and $Q = sh + wK$ in a phase-coherent kernel-economical configuration. Then physical nonnegativity is equivalent to*

$$pw - y \leq s \leq x - qw, \quad (36)$$

with

$$0 \leq w \leq \left\lfloor \frac{x + y}{r} \right\rfloor. \quad (37)$$

At fixed w , the number of integer s -positions is

$$x + y - rw + 1. \quad (38)$$

Proof. Since

$$P = \mu - Q = (x, y) - s(1, -1) - w(q, p),$$

one has

$$P_0 = x - s - qw, \quad P_1 = y + s - pw.$$

Both are nonnegative exactly when (36) holds. The interval is nonempty exactly when

$$rw \leq x + y,$$

giving (37). Its number of integer points is the upper endpoint minus the lower endpoint plus one, yielding (38). $\square$

8 Semantic uniqueness and physical lift freedom

The previous sections show that the semantic trajectory is input-determined while the physical representation can leave the canonical section. We now state this as a quotient structure.

Definition 8.1 (Semantic fiber). *For $m \in \mathcal{M}_{p,q}$, define*

$$F_m = \Theta^{-1}(m).$$

Two rich configurations are semantically equivalent if

$$c \sim c' \iff \Theta(c) = \Theta(c').$$

Theorem 8.2 (Fiber congruence and canonical semantic quotient). *In every exact literal-demand lift, $\sim$ is preserved by the input transitions. Consequently, the quotient transition system $\mathcal{C}/\sim$ is isomorphic to the canonical minimum-demand dynamics $\mathcal{M}_{p,q}$.*

Proof. If $\Theta(c) = \Theta(c')$, then exactness (9) gives

$$\Theta(\delta_a(c)) = U_a(\Theta(c)) = U_a(\Theta(c')) = \Theta(\delta_a(c')).$$

Thus $\sim$ is a transition congruence. The map sending a fiber to its common semantic value is a well-defined bijection from the quotient onto $\mathcal{M}_{p,q}$, and it intertwines the transitions. $\square$

Definition 8.3 (Canonical retraction). *Define*

$$N = \sigma \circ \Theta. \quad (39)$$

Proposition 8.4 (Canonical absorption). *The map N is idempotent,*

$$N^2 = N,$$

and for every input word w ,

$$N \circ \delta_w = N \circ \delta_w \circ N. \quad (40)$$

Proof. Since $\Theta \circ \sigma = \text{id}$,

$$N^2 = \sigma \Theta \sigma \Theta = \sigma \Theta = N.$$

For one symbol,

$$N \delta_a = \sigma \Theta \delta_a = \sigma U_a \Theta,$$

while

$$N \delta_a N = \sigma \Theta \delta_a \sigma \Theta = \sigma U_a \Theta \sigma \Theta = \sigma U_a \Theta.$$

Induction on the length of w yields (40). $\square$

Remark 8.5 (What is unique). *The residual, minimum demand, kernel level, semantic carry indicator, and canonical quotient path are unique. These are semantic objects.*

Proposition 8.6 (Unrestricted physical lift-presentation non-uniqueness). *The canonical semantic quotient does not determine a unique physical deterministic pushdown presentation.*

Proof. Start from any exact deterministic pushdown presentation with finite control set Q . For an integer $m \geq 2$, adjoin an independent cyclic control coordinate $j \in \mathbb{Z}_m$, update it by $j \mapsto j + 1 \pmod{m}$ on every consumed input symbol, and let the stack dynamics, semantic projection, and acceptance test ignore this coordinate. The resulting machine is again an exact deterministic lift and has control set $Q \times \mathbb{Z}_m$.

Reachability of the section can be made explicit. Because the original section lies in the reachable configuration set, for each semantic state s choose a word u_s reaching $\sigma(s)$; use $(\sigma(s), |u_s| \pmod{m})$ as the decorated section point. Moreover, when $\gcd(m, p + q) = 1$, appending powers of the balanced word $0^q 1^p$ preserves the semantic residual while advancing the cyclic coordinate by $p + q$, so every cyclic sheet is reachable in the corresponding semantic fiber. Taking infinitely many pairwise distinct primes $m \nmid (p + q)$ gives pairwise non-isomorphic finite-control presentations (their control-set cardinalities are $|Q|m$) with the same semantic quotient. Hence physical presentation uniqueness fails without an explicit reduction/minimality category. $\square$

Remark 8.7 (Correct uniqueness statement). *The conclusion of this paper is therefore*

$$\boxed{\text{unique semantic core; non-unique physical normalization lift.}}$$

The stronger problem of classifying minimal or section-preserving lifts is deliberately deferred. In particular, canonical endpoint coherence is not presented as a confluence theorem.

9 The historical 2:1 machine as a specialization

Set

$$p = 2, \quad q = 1, \quad r = 3, \quad K = (1, 2).$$

The companion construction/geometry analyses of the historical recognition core yield the state-potential invariant [7, 8]

$$\rho = D(S) + \Phi(q_{\text{state}}), \quad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3, \quad (41)$$

where

$$D(S) = 2N_0(S) - N_1(S).$$

The present theory identifies this potential with the scalar quantity-defect projection.

Corollary 9.1 (Historical states as SEARCH sheets). *In the phase-coherent quantity shadow,*

$$\Phi = rs = 3s.$$

Hence

$$s(C) = 0, \quad s(L) = 1, \quad s(R) = -1.$$

Proof. Equation (41) says that the state potential is the difference between the semantic residual and the physical literal residual. This is precisely $\Phi(Q)$. Corollary 6.3 gives $\Phi(Q) = 3s$. $\square$

The historical input-time literal stack changes are

$$-1, +2, +5.$$

The established reachable-store families for this historical core,

$$C : 0^* \cup 1^* \cup \{10\}, \quad L : 0^*, \quad R : 1^* \cup \{0\},$$

are kernel-economical under the state-potential invariant above. Moreover, all three input-time length changes are congruent to $-1 \pmod{3}$; starting from the empty logical stack this gives $|P_t| \equiv -t \pmod{3}$, which matches the semantic demand clock. Thus the historical input-time run lies in the phase-coherent, kernel-economical class required by the materialization results below.

Corollary 9.2 (Historical stack actions as materialization bursts). *The physical materialization amounts are*

$$-1 \longleftrightarrow M = 0, \quad +2 \longleftrightarrow M = 1, \quad +5 \longleftrightarrow M = 2.$$

Thus the historical +5 action is a two-kernel-packet burst materialization.

Proof. Apply (26) with $r = 3$. $\square$

Corollary 9.3 (Structural derivation of $b + 2c = n$). *Let b be the number of historical +2 transitions and c the number of historical +5 transitions in an accepted 2:1 computation with $N_0 = n$ and $N_1 = 2n$. Then*

$$b + 2c = n. \tag{42}$$

Proof. By Corollary 9.2, each +2 transition contributes one unit to the total materialization budget and each +5 contributes two; -1 transitions contribute zero. Theorem 7.11 gives total budget n . $\square$

Remark 9.4 (Eager/lazy reconstruction). *A companion construction study shows that the direct centered 2:1 specialization and the historical transition table differ in two reachable right-side matching families. Replacing the lazy successor in those families by canonical eager materialization recovers the historical table. In the present terminology, this is normalization chronology freedom: the semantic carry schedule is fixed, while the physical materialization schedule need not be.*

10 Discussion and scope boundary

The theory developed above has two levels that must not be conflated.

Universal semantic layer. The kernel-level formula, minimum demand, semantic update factorization, semantic WRITE count, and semantic chronology are properties of the weighted balance arithmetic itself. They do not depend on a particular stack representation.

General literal-LIFO layer. The top-rewrite suffix criterion and the obstruction classification depend only on one-top real-time locality and the fixed canonical literal orientation. They reveal that noncanonical representation is sometimes forced. The resulting debt need not be a quantity defect: carry-order obstruction gives a zero-quantity but nonzero-order example.

Phase-coherent WSC count-shadow layer. The lattice $\mathcal{D}_{p,q}$, the (s, w) coordinates, physical materialization level λ , backlog law, and factorization (30) belong to the explicitly scoped phase-coherent count-shadow class. Kernel economy is additionally used when w is interpreted as a nonnegative backlog and when eager/lazy dichotomies or global budget arguments require exclusion of redundant kernel workspace.

Uniqueness layer. The semantic quotient is unique in every exact lift, but physical lifts are not. This paper does not classify all possible order-debt policies, control-debt policies, or minimal lifts. Nor does it claim that all realizations are connected by local SEARCH/WRITE commutations. Those questions form the natural next stage.

10.1 Claims deliberately excluded from v1.5

For clarity, the following statements are *not* claimed:

1. Every correct DPDA is phase-coherent or WSC-factorizable.
2. Every noncanonical configuration has nonzero quantity defect.
3. Quantity defect alone characterizes canonicity.
4. Vector quantity defect is constant on a control state.
5. Every LIFO obstruction forces exactly $\pm(1, -1)$; that conclusion is proved for the buried-CANCEL family under kernel economy.
6. Every transition is syntactically exactly one of SEARCH, CANCEL, or WRITE.
7. Semantic WRITE timing is non-unique. It is unique; physical materialization timing is the non-unique chronology.
8. Unique canonical endpoints imply confluence.
9. All admissible physical lifts are equivalent under local commuting moves.
10. The historical transition table is the unique realization of the 2:1 semantics.
11. The WSC lattice is the full representation fiber.

11 Conclusion

The weighted balance problem exposes a structural separation that is easy to miss when one reads a pushdown transition table only operationally. The shortest completion demand is a unique semantic object. Its evolution is governed by a primitive kernel packet and factors into a unit cancellation plus an optional kernel carry. Real-time one-top locality then determines

whether that semantic update can be realized directly in canonical literal form. When it cannot, the machine must leave the canonical section and carry normalization debt.

The debt itself has several possible channels. Quantity debt is only one. Stack order and finite-control sheets can carry information that the Parikh count shadow forgets. Within the phase-coherent WSC count-shadow class, however, the arithmetic becomes rigid: defects lie in a two-generator lattice, physical stack changes determine a unique materialization amount, and every transition admits the factorization

$$\Delta P = -e_a - \alpha(1, -1) + M(q, p).$$

This explains why CANCEL, SEARCH, and WRITE recur as structural roles rather than isolated transition-table tricks.

The historical 2:1 automaton provides a concrete validation of the framework. Its side-state potentials become SEARCH-sheet coordinates, its mod-3 stack clock becomes the $p + q$ phase clock, its +5 transition becomes a two-packet materialization burst, and its earlier identity $b + 2c = n$ becomes a direct budget law.

The semantic core is unique; physical normalization lifts are not. The next natural problem is therefore not to seek another construction, but to classify the lift freedom: which branching is forced by LIFO obstruction, which branching is redundant, and under what minimality conditions can two realizations be regarded as the same lift? That question leads from the present WSC structural theory toward a broader theory of real-time pushdown normalization.

A Theorem dependency map

The logical dependency structure of the paper can be summarized as follows.

- **Semantic arithmetic chain:**

Kernel-fiber lemma $\Rightarrow$ Dominating kernel level $\Rightarrow$ semantic update factorization.

This yields semantic WRITE count, the demand-length clock, and semantic chronology uniqueness.

- **LIFO locality chain:**

canonical literal section + one-top locality $\Rightarrow$ top-rewrite suffix criterion.

The suffix criterion yields the buried-CANCEL and carry-order obstruction families. With kernel economy, the buried-CANCEL family yields forced unit SEARCH; the carry-order family yields the minimum order-debt bound. Three-channel canonicity together with explicit quantity-, order-, and control-only witnesses yields configuration-level channel independence.

- **Exact-lift chain:**

exact lift semantics $\Rightarrow$ universal representation conservation
and independently

exact lift semantics $\Rightarrow$ fiber congruence / semantic quotient
 $\Rightarrow$ canonical retraction / absorption.

- **Phase-coherent WSC chain:**

phase coherence $\Rightarrow$ WSC defect lattice $\Rightarrow (s, w)$ coordinates $\Rightarrow \lambda \Rightarrow M$.

This yields the physical WSC factorization and the mod- $(p + q)$ physical clock. Adding kernel economy gives the eager/lazy dichotomy and the global materialization budget.

B Result status and paper boundary

Result	Status in v1.5	Scope / note
Kernel-fiber lemma	Proved	Arithmetic, realization-independent
Dominating kernel-level formula	Proved	Arithmetic, realization-independent
Semantic CANCEL/carry factorization	Proved	Arithmetic, realization-independent
Semantic chronology uniqueness	Proved	Arithmetic, realization-independent
Top-rewrite suffix criterion	Proved	One-top real-time locality
Direct obstruction classification	Proved	Fixed orientation 1^*0^*
Forced unit SEARCH	Proved	Buried CANCEL + kernel economy
Minimum carry-order debt	Proved	Count-exact repair in carry-order family
Quantity/order/control debt separation and independence	Proved	Full canonicity refinement; configuration-level witnesses
Universal representation conservation	Proved	Exact lift, count shadow
WSC defect lattice	Proved	Phase-coherent class
Physical WSC factorization	Proved	Phase-coherent one-top class
Eager/lazy dichotomy	Proved	Canonical quantity state + economy
Global materialization budget	Proved	Accepted run + economy
Semantic quotient uniqueness	Proved	Exact-lift model
Physical lift-presentation non-uniqueness	Proved	No minimality restriction
Historical 2:1 state-sheet interpretation	Corollary	Uses established historical potential invariant
Historical $-1, +2, +5$ materialization reading	Corollary	$r = 3$ specialization
$b + 2c = n$ structural derivation	Corollary	Accepted historical 2:1 computation
Representation-relative reduced-lift theory	Deferred here	Developed in the subsequent uniqueness/lift companion; unrestricted classification remains open
Section-preserving lift morphisms	Deferred here	Developed in the subsequent uniqueness/lift companion
Full confluence / critical-pair theory	Deferred	Not implied by endpoint uniqueness
Higher-rank pushdown normalization	Deferred	Long-term general theory

C Executable consistency audit

A separate script, `wsc_validation_v1_1.py`, serves as a regression check for the symbolic arguments. It is not used as a substitute for proof. The current audit exhausts all coprime $1 \leq p, q \leq 8$ and all binary prefixes of length at most nine for the closed-form minimum-demand

formula, semantic carry criterion, and demand clock; it also checks the complete one-top direct-realizability classification on bounded canonical demand regions, the defect-lattice coordinate formulas, the count-shadow fiber inequalities, the physical-factorization reconstruction formulas, and bounded exhaustive instances of the minimum carry-order debt. The current run performs 477,033 assertion-level checks and passes.

D Roadmap beyond this paper

A subsequent companion manuscript develops semantic uniqueness and LIFO lift theory beyond the exact WSC count-shadow class, including machine-independent semantic quotients, representation-relative reduction, obstruction-forced fiber branching, section-preserving morphisms, and normalization-schedule structure [6]. That companion deliberately keeps unrestricted DPDA lift minimality and classical confluence open. Together, the present structural theory and the companion lift theory are intended as the two principal inputs to a later *General Theory of Real-Time Pushdown Normalization*.

Statements and Declarations

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Code and validation artifacts. Validation scripts and checkpoint material are maintained in the author’s public research repositories. The executable checks are supplementary consistency evidence and do not replace the symbolic proofs.

References

- [1] R. Alur, V. Kumar, P. Madhusudan, and M. Viswanathan. Congruences for visibly pushdown languages. In *Automata, Languages and Programming (ICALP 2005)*, LNCS 3580, pp. 1102–1114, 2005. doi:10.1007/11523468_89.
- [2] O. Gauwin, A. Muscholl, and M. Raskin. Minimization of visibly pushdown automata is NP-complete. *Logical Methods in Computer Science*, 16(1):14, 2020. doi:10.23638/LMCS-16(1:14)2020.
- [3] G. Sénizergues. $L(A) = L(B)$? Decidability results from complete formal systems. *Theoretical Computer Science*, 251(1–2):1–166, 2001. doi:10.1016/S0304-3975(00)00285-1.
- [4] P. Jančar. A short decidability proof for DPDA language equivalence via first-order grammars. CoRR abs/1010.4760, 2011.
- [5] A. Carayol and O. Serre. Marking shortest paths on pushdown graphs does not preserve MSO decidability. *Information Processing Letters*, 116(10):638–643, 2016. doi:10.1016/j.ipl.2016.04.015.
- [6] A. E. Bütün. *Semantic Uniqueness and LIFO Lift Theory for Real-Time Pushdown Normalization*. Companion manuscript, 2026.
- [7] A. E. Bütün. *Exact Structure and Resource Bounds in Deterministic Pushdown Automata for Binary Counting Languages*. Manuscript, 2026.

- [8] A. E. Bütün. *Real-Time $p:q$ DPDA Generalization*. Manuscript, 2026.